

# SANNIHITHA GUDIMALLA

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## SUMMARY

Innovative Software Engineer specializing in product development with 2+ years of experience designing, building, testing, and optimizing large-scale infrastructure applications, web applications, and distributed systems. Proven success in reducing transaction latency by 20%, improving system stability by 15%, and increasing automated test pass rates by 40%. Proficient in industry-standard languages (C, C++, Java, C#) and modern development technologies (Python, JavaScript, SQL, HTML, CSS, REST APIs). Skilled in object-oriented design, multi-threaded programming, and relational databases. Adept at collaborating in Agile environments, applying CI/CD practices, and producing high-quality, well-tested code aligned with business and product goals.

## EDUCATION

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|------------------------------------------------------------------------------|---------------------|
| Illinois Institute of Technology, Chicago, IL                                | JAN 2024 – DEC 2025 |
| <b>Masters in computer science</b> , GPA 3.44 /4.00                          |                     |
| VNR Vignana Jyothi Institute of Engineering and Technology, Hyderabad, India | JUN 2019 – JUL 2022 |
| <b>Bachelor of Technology, Computer Science</b> , GPA 3.7 /4.00              |                     |

## SKILLS

- **Programming Languages:** C++, C#, Java, Python, JavaScript
- **Web Development:** HTML, CSS, REST APIs, Node.js, Spring Boot, Qt, PHP (basic)
- **Databases:** MySQL, MongoDB, SQL (queries, optimization)
- **Frameworks & Tools:** Spring Boot, Node.js, Qt, Docker, Jenkins, Git, GraphQL
- **Cloud & DevOps:** AWS, Google Cloud Platform (GCP)
- **System Design:** Large-scale infrastructure, Distributed Systems, Object-Oriented Design, Multi-threading, Functional Requirements Analysis
- **Testing & Deployment:** CI/CD, Automated Testing, Unit & Integration Testing
- **Data Analysis & ML:** Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, Keras
- **Operating Systems:** UNIX/Linux (Shell scripting, system monitoring, networking tools)

## EXPERIENCE

### SOFTWARE ENGINEER I

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|-----------------------------|----------------------|
| NCR Voyix, Hyderabad, India | JULY 2022 - DEC 2023 |
|-----------------------------|----------------------|

- Engineered and maintained the Promotion Engine (PE), a large-scale infrastructure system processing millions of daily retail transactions at both store and enterprise levels, using C++ (Qt) and Java (Spring Boot).
- Optimized and extended web applications with REST APIs, improving data flow efficiency and reducing average transaction processing time by 20%, enhancing checkout experience for enterprise clients.
- Automated CI/CD pipelines via Jenkins, increasing automated test pass rates by 40%, accelerating release cycles, and minimizing production regression bugs.
- Diagnosed and resolved multi-threaded concurrency issues and memory leaks, improving system stability by 15% and reducing customer-reported errors.
- Integrated and optimized relational database (MySQL) queries to enhance performance and reduce data retrieval latency.
- Leveraged Google Cloud Platform (GCP) for scalable, reliable deployment and management of production applications.
- Actively participated in Agile sprint planning, retrospectives, and cross-functional strategy sessions to align engineering goals with broader business objectives.

### SOFTWARE INTERN

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|-----------------------------|----------------------|
| NCR Voyix, Hyderabad, India | JAN 2022 - JUNE 2022 |
|-----------------------------|----------------------|

- Contributed to the development of distributed systems for the Promotion Engine using C++, Java, and REST APIs, improving system modularity and scalability.
- Developed and refined web interfaces using HTML, CSS, and JavaScript to enhance internal tool usability.
- Gained hands-on experience in **cloud deployment** via GCP and relational database integration.

## PROJECTS

- **Handwriting Recognition using CRNN + CTC, TrOCR**  
Started with a CRNN model (CNN + BiLSTM + CTC) in TensorFlow for handwritten word recognition on the IAM dataset but faced accuracy limitations. Switched to Hugging Face's TrOCR, achieving 80% accuracy through optimized preprocessing, batch decoding, and evaluation using Character Error Rate (CER).
- **Spammer Detection & Fake User Identification on Twitter**  
Developed a machine learning system using SVM, Naïve Bayes, and Random Forest to detect fake accounts, spam tweets, and malicious URLs. Engineered features and applied PCA for dimensionality reduction, achieving high classification accuracy on real Twitter data.
- **Crypto Trends Analyzer – Bitcoin Price Forecasting & Portfolio Optimization**  
Built ML models (Random Forest, XGBoost, AdaBoost) using historical and sentimental data to predict Bitcoin price movements and trends. Performed advanced feature engineering, dimensionality reduction (PCA, t-SNE).

## CERTIFICATIONS

Full Stack Development (Udemy) | Data Analysis with Python (Coursera) | Cloud Computing (Coursera)